Valuation Analysis of ASML Holding N.V.

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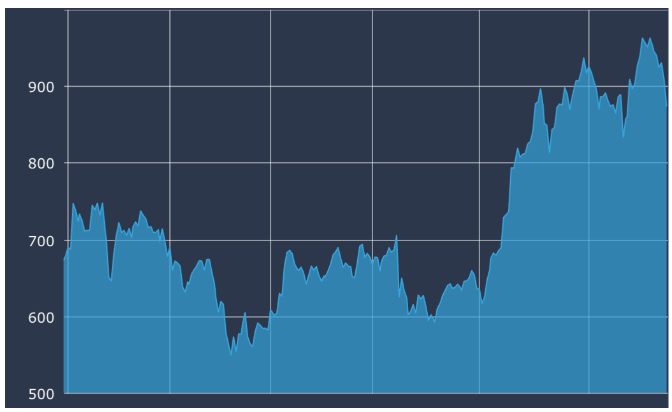
December 2025

Valuation Overview

Our discounted cash flow valuation (based on the Gordon growth model) of ASML Holding N.V. as of 31-12-2024 found the value of the stock to be mildly overvalued (~14%) in street conditions from its pre-earnings market price, largely justified due to the markets expectation of ASML holding a monopoly over the EUV lithography system segment, and its domination in the DUV segment.

As of its conditions of 31-12-2024, a hold decision would be advisable, as ASML appears close to fairly valued, with limited near-term upside, but still strong potential for growth driven by future adoption of its high-NA EUV systems.



The stock ended up falling to 550 euros after FY24 earnings were announced, bringing the market value closer to the intrinsic value predicted by the DCF model. The decline appears to not be driven by the earnings report but rather by uncertainty and tensions regarding exports and customer deferrals (notably from Chinese players).

Assumptions

The DCF model was heavily based on a scenario analysis with multiple drivers. Three scenarios (conservative, street, and optimistic) were applied to major line items in the income statement like revenue, gross profit, EBITDA, and D&A. The street case of these major line items followed a mean regression assumption with suitable adjustments applied to match with an EUV scaling cycle, followed by a decrease to finally normalize.

A terminal growth rate of 2.00% for conservative cases, 2.50% for the street case, and 3.00% for an optimistic case was applied. This terminal growth rate aligns with long term Euro-zone nominal GDP growth rates, along with reflecting ASML's strategic market position. The model is most sensitive to changes in EBITDA margin, weighted average cost of capital, and terminal growth rate assumptions. The effect of EBITDA margin is explained as EBITDA directly affects the free cash flow throughout the forecast period.

WACC Calculations

Weighted Average Cost of Capital		
Terms		
WACC = Ke*(E/E+D)+Kd*(D/E+D)*(1-tax)		8.0459%
Ke = Rf + Beta*(Rm-Rf)		8.1376%
Rf		2.86%
Rf-Rm (Market Equity Risk Premium)		4.33%
Beta		1.22
Kd		3.47%
E	€	2,66,92,16,60,764
D	€	4,68,76,00,000
%E		98.27%
%D		1.73%
Tax Rate		18.59%

- We arrived at a WACC of 8.05%
- A risk free rate of 2.86% was chosen which aligns with the Euro Area Government 10Y bond rate.
- The market equity risk premium of 4.33% was selected which aligns with Damodaran's viewpoint as of Jan 2025.
- A beta of 1.36 which underwent an adjustment according to Bloomberg's Mean Reversion assumptions, led to a beta of 1.22.
- The equity weight was calculated to be 98.27% and the debt weight was 1.73%.
- A tax rate of 18.59% was chosen, the average of the past 5 years of ASMLs operations.
- A conservative case WACC of 8.45% aligning with a systematic EUV downplay and an optimistic case WACC of 7.64% were used, aligning with a heavy EUV adoption, driven by AI scaling.

Scenario Analysis

A conservative viewpoint values ASML Holding N.V. at 336 euros, indicating a heavy 50% overvaluation of the stock price. The conservative case assumes a long term semiconductor downcycle, and extended export restrictions on ASML, leading to a systematic decrease in shipment volumes. A heavy conservative estimation of key drivers like revenue growth (40% decrease of street case), gross profit margin and EBITDA margin (both having a 5% decrease of street case), lead to this low intrinsic valuation.

An optimistic viewpoint values ASML Holding N.V. at 887 euros, a 31% undervaluation of the stock price. We buy the company projections of revenue growth in this case (20% increase of street case), and we hence also increase EBITDA margin and gross profit by 5% and 7.5% of the street case respectively.